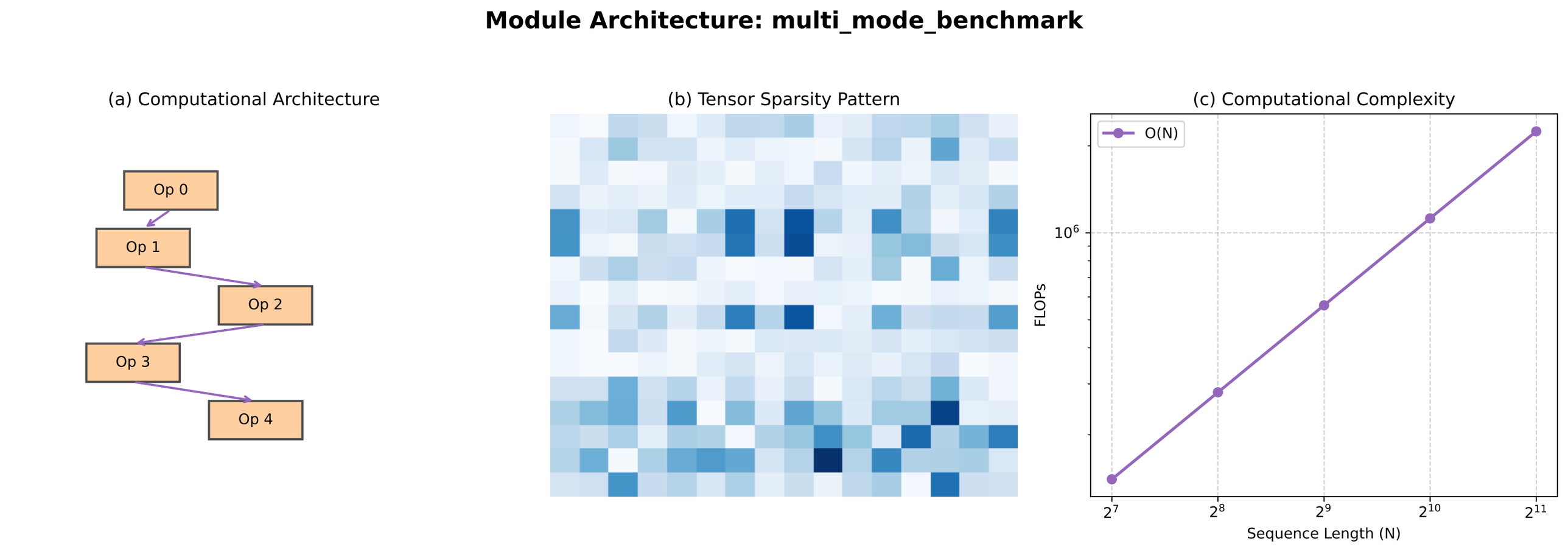Module Architecture: multi_mode_benchmark